A chain rule for the derivative of containers under composition

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Abstract

We prove, at the level of containers over **Set**, the chain rule

$$\partial(G \lhd F) \;\cong\; (\partial G \lhd F) \times \partial F,$$

where $\lhd$ is container composition (substitution), ∂ is the one-hole-context derivative of Abbott–Altenkirch–Ghani–McBride, and $\times$ is the binary product of containers. The proof is a single explicit isomorphism whose content is a case split on whether a chosen position lies in the differentiated outer block or one of the untouched inner blocks. We first dispatch the atoms and the Leibniz rule, all instances of the same one-move template. The result is the container shadow of the Faà di Bruno / analytic-functor chain rule; with composition $\lhd$ as the substitution operation it does not appear in AAGM 2003, and is original in the classical container setting (cf. Joram–Veltri arXiv:2512.17484 for the univalent case). Verified computationally on 23 finite containers (explicit shape and per-shape position bijections).

1 Setup

We work in **Set** with classical logic; the one place this matters is Lemma 1 and is flagged there.

Definition 1 (Container, extension)**.** A *container* $F = S \lhd P$ is a set of *shapes* S together with, for each $s : S$, a set of *positions* $P\,s$. Its *extension* is the functor $[\![S \lhd P]\!] : \mathbf{Set} \to \mathbf{Set}$,

$$[\![S \lhd P]\!]\,X \;=\; \textstyle\sum_{s:S} (P\,s \to X).$$

A *morphism* $(S \lhd P) \to (S' \lhd P')$ is a pair (u, f) with $u : S \to S'$ and $f : \prod_{s:S} P'(u\,s) \to P\,s$ (positions run contravariantly). Containers and their morphisms form the category **Cont**; $[\![-]\!]$: $\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is fully faithful. An *isomorphism* of containers is thus a bijection u of shapes together with, for each s, a bijection $P'(u\,s) \cong P\,s$.

Definition 2 (Operations on **Cont**)**.** For $F = S \lhd P$ and $G = T \lhd Q$:

- *Sum.* $F + G = (S + T) \lhd [P, Q]$, with positions inherited: $[P, Q](\mathrm{inl}\,s) = P\,s$, $[P, Q](\mathrm{inr}\,t) = Q\,t$. Extension $[\![F]\!] + [\![G]\!]$.
- *Product.* $F \times G = (S \times T) \lhd \lambda(s, t).\, P\,s + Q\,t$. Extension $[\![F]\!] \times [\![G]\!]$.
- *Composition* (substitution) $G \lhd F$, with $[\![G \lhd F]\!] = [\![G]\!] \circ [\![F]\!]$:

$$\text{shapes} \;\; \textstyle\sum_{t:T} (Q\,t \to S), \qquad \text{positions at } (t, f) \;\; \textstyle\sum_{q:Q\,t} P(f\,q).$$

The constants are $0 = \varnothing \lhd !$ (no shapes), $1 = \mathbf{1} \lhd (\lambda\_.\, \varnothing)$ (one shape, no positions; $[\![1]\!]$ = constant $\mathbf{1}$), and the identity $\mathrm{Id} = \mathbf{1} \lhd (\lambda\_.\, \mathbf{1})$ (one shape, one position; $[\![\mathrm{Id}]\!] = \mathrm{Id}_{\mathbf{Set}}$). For a set A, $K_A = A \lhd (\lambda\_.\, \varnothing)$.

Definition 3 (Derivative)**.** The *derivative* of $F = S \lhd P$ is

$$\partial F \;=\; \Big(\textstyle\sum_{s:S} P\,s\Big) \;\lhd\; \lambda(s, p_0).\, \{\, p : P\,s \mid p \neq p_0 \,\}.$$

A shape of ∂F is a *pointed* shape (s, p_0) — a shape with a chosen hole p_0 — and its positions are the *other* positions. This is the one-hole-context functor of Abbott–Altenkirch–Ghani–McBride ("Derivatives of Containers", 2003/2005); $[\![\partial F]\!]\,X$ is the set of one-hole F-structures over X.

Throughout, for a set B and a point $b_0 : B$ write $B \setminus b_0 := \{b : B \mid b \neq b_0\}$.

2 The atoms and the Leibniz rule

Every identity below is one move: *a hole in a structured position-set is a choice of which summand the hole lives in, plus a hole there.* We record them because the chain rule is the same move applied to a *dependent* sum of position-sets.

Proposition 1. $\partial\,\mathrm{Id} \cong 1, \quad \partial K_A \cong 0$ *(in particular $\partial 1 \cong 0$).*

Proof. $\partial\,\mathrm{Id}$ has shapes $\sum_{s:\mathbf{1}} \mathbf{1} \cong \mathbf{1}$, and at that shape positions $\{p : \mathbf{1} \mid p \neq p_0\} = \varnothing$; so $\partial\,\mathrm{Id} = \mathbf{1} \lhd (\lambda\_.\, \varnothing) = 1$. ∂K_A has shapes $\sum_{a:A} \varnothing = \varnothing$, hence equals 0. $\square$

Proposition 2 (Sum rule)**.** $\partial(F + G) \cong \partial F + \partial G$.

Proof. Shapes of $\partial(F+G)$ are $\sum_{u:S+T} [P, Q]\,u \;=\; \sum_{s:S} P\,s \;+\; \sum_{t:T} Q\,t$, i.e. shapes of ∂F plus shapes of ∂G, by the universal property of $+$. At $\mathrm{inl}(s, p_0)$ the positions are $\{p : P\,s \mid p \neq p_0\}$, exactly the ∂F positions; symmetrically at $\mathrm{inr}(t, q_0)$. The bijections are identities. $\square$

Proposition 3 (Leibniz / product rule)**.** $\partial(F \times G) \cong (\partial F \times G) + (F \times \partial G)$.

Proof. A shape of $\partial(F \times G)$ is $\big((s,t),\, h\big)$ with hole $h : P\,s + Q\,t$. The coproduct splits the hole:

$$\textstyle\sum_{(s,t)} (P\,s + Q\,t) \;\cong\; \Big(\sum_{(s,t)} P\,s\Big) + \Big(\sum_{(s,t)} Q\,t\Big) = \mathrm{sh}(\partial F \times G) + \mathrm{sh}(F \times \partial G).$$

If $h = \mathrm{inl}\,p_0$, the positions $\{x : P\,s + Q\,t \mid x \neq \mathrm{inl}\,p_0\}$ are $(P\,s \setminus p_0) + Q\,t$, exactly the positions of $\partial F \times G$ at $((s, p_0), t)$. If $h = \mathrm{inr}\,q_0$, they are $P\,s + (Q\,t \setminus q_0)$, the positions of $F \times \partial G$ at $(s, (t, q_0))$. $\square$

3 The chain rule

The one ingredient beyond the atoms is the decomposition of a function out of a pointed domain.

Lemma 1 (Pointed-domain splitting)**.** *Let B be a set and $b_0 : B$. The map $\mathbf{1} + (B \setminus b_0) \to B$, $\mathrm{inl}\,\star \mapsto b_0$, $\mathrm{inr}\,b \mapsto b$, is a bijection. Consequently, for any set S,*

$$(B \to S) \;\xrightarrow{\;\cong\;}\; S \times \big((B \setminus b_0) \to S\big), \qquad f \mapsto \big(f\,b_0,\; f|_{B \setminus b_0}\big).$$

Proof. Injectivity of the first map: the images of inl and inr are disjoint because $b_0 \notin B \setminus b_0$, and each restriction is injective. Surjectivity: every $b : B$ satisfies $b = b_0$ or $b \neq b_0$ (excluded middle on the proposition $b = b_0$), landing in the image of inl resp. inr. Applying $(- \to S)$ to $B \cong \mathbf{1} + (B \setminus b_0)$ and using $(\mathbf{1} + C \to S) \cong S \times (C \to S)$ gives the second statement. $\square$

Remark 1. Surjectivity is the only use of classical logic in this note. Over a general base it requires b_0 to be a *decidable* (complemented) point of B; over **Set** this is automatic. This is precisely the hinge that Joram–Veltri (arXiv:2512.17484) must handle with care in univalent foundations, where $B \setminus b_0$ is the type $\sum_{b:B}(b \neq b_0)$ and the splitting holds after set-truncation. For directed containers over **Set** it is free.

Theorem 1 (Container chain rule)**.** *For containers $F = S \triangleleft P$ and $G = T \triangleleft Q$,*

$$\partial(G \triangleleft F) \;\cong\; (\partial G \triangleleft F) \times \partial F$$

in **Cont***.*

Proof. We exhibit a shape bijection and, over it, a position bijection.

Both sides explicitly. By Definitions 2–3:

Left. $G \triangleleft F$ has shapes $\sum_{t:T}(Q\,t \to S)$ and positions $R(t,f) = \sum_{q:Q t} P(f\,q)$. Hence a shape of $\partial(G \triangleleft F)$ is

$$(t,\ f,\ q_0,\ p_0), \qquad t : T,\ f : Q\,t \to S,\ q_0 : Q\,t,\ p_0 : P(f\,q_0),$$

(the hole is the position $(q_0, p_0) \in R(t,f)$), with positions

$$\mathrm{pos}_L = \big\{(q,p) : \textstyle\sum_{q:Qt} P(f\,q) \;\big|\; \neg(q = q_0 \wedge p = p_0)\big\}. \tag{1}$$

Right. ∂G has shapes (t, q_0) $(t : T,\ q_0 : Q\,t)$ and positions $Q\,t \setminus q_0$. So $\partial G \triangleleft F$ has shapes $\big(t, q_0,\ g\big)$ with $g : (Q\,t \setminus q_0) \to S$, and positions $\sum_{q':Qt\setminus q_0} P(g\,q')$. And ∂F has shapes (s_*, p_*) with positions $P\,s_* \setminus p_*$. Thus a shape of the right-hand product is

$$\big((t, q_0, g),\ (s_*, p_*)\big),$$

with positions, by the product rule of Definition 2,

$$\mathrm{pos}_R = \Big(\textstyle\sum_{q':Qt\setminus q_0} P(g\,q')\Big) \;+\; \big(P\,s_* \setminus p_*\big). \tag{2}$$

Shape bijection φ. Send

$$\varphi\,(t, f, q_0, p_0) \;=\; \big((t,\ q_0,\ f|_{Qt\setminus q_0}),\ (f\,q_0,\ p_0)\big).$$

Holding (t, q_0) fixed, this is exactly the map $f \mapsto (f\,q_0,\ f|_{Qt\setminus q_0})$ paired with the identity on p_0, which is a bijection by Lemma 1 (here $B = Q\,t$, $b_0 = q_0$): giving $f : Q\,t \to S$ is the same as giving $s_* := f\,q_0 \in S$ and $g := f|_{Qt\setminus q_0}$, and then $p_0 : P(f\,q_0) = P\,s_*$ is the data p_*. Summing the bijection over all (t, q_0) gives a bijection of shape sets.

Position bijection. Fix a left shape (t, f, q_0, p_0) and its image, so $s_* = f\,q_0$, $g = f|_{Qt\setminus q_0}$, $p_* = p_0$. Define $\psi : \mathrm{pos}_L \to \mathrm{pos}_R$ on (1) by the hole-case split:

$$\psi(q,p) = \begin{cases} \mathrm{inl}(q,p) & q \neq q_0, \\ \mathrm{inr}(p) & q = q_0. \end{cases}$$

Well-defined. If $q \neq q_0$ then $p : P(f\,q) = P(g\,q)$, so (q,p) is a valid element of the first summand of (2). If $q = q_0$ then $p : P(f\,q_0) = P\,s_*$, and the constraint $\neg(q = q_0 \wedge p = p_0)$ forces $p \neq p_0 = p_*$, so $p \in P\,s_* \setminus p_*$, the second summand.

Bijective. The two cases hit inl and inr disjointly, so it suffices to check each is a bijection onto its summand. For the first: $q \mapsto q$, $p \mapsto p$ with $P(f\,q) = P(g\,q)$ is the identity on $\sum_{q\neq q_0} P(f\,q) = \sum_{q'\in Qt\setminus q_0} P(g\,q')$. For the second: $p \mapsto p$ is the identity $P(f\,q_0) \setminus p_0 = P\,s_* \setminus p_*$. The inverse sends $\mathrm{inl}(q', p') \mapsto (q', p')$ (legal since $q' \neq q_0$) and $\mathrm{inr}(p') \mapsto (q_0, p')$ (legal since $p' \neq p_* = p_0$).

(φ, ψ) is a bijection of shapes together with a bijection of positions over each shape, i.e. an isomorphism in **Cont**. $\square$

Where the two factors come from. The composite's position-set is the *dependent* (indexed) coproduct $\coprod_{q:Qt} P(f\,q)$ over the G-positions $Q\,t$. Differentiating it — choosing a hole — means choosing an index q_0 *and* a hole inside that block. The leftover splits as

$$\underbrace{\coprod_{q \neq q_0} P(f\,q)}_{\text{other blocks: } \partial G \triangleleft F} \quad + \underbrace{\big(P(f\,q_0) \setminus p_0\big)}_{\text{this block: } \partial F}.$$

Choosing q_0 and keeping the other blocks full is exactly ∂G applied to the $Q\,t$ index with the blocks substituted by F — that is $\partial G \triangleleft F$ — while the punctured block at q_0 is exactly ∂F. Their juxtaposition is the product (positions add). The chain rule is Leibniz for an *indexed* product.

4 Sanity checks

Corollary 1 (The y^2 case fixes the convention)**.** *Take $G = y^2$, i.e. $T = \mathbf{1}$, $Q\star = \mathbf{2}$. Then $G \triangleleft F \cong F \times F$, and the theorem reads $\partial(F \times F) \cong (\partial(y^2) \triangleleft F) \times \partial F$. Since $\partial(y^2) = \mathbf{2} \triangleleft \mathbf{1} = y + y$, we get $\partial(y^2) \triangleleft F \cong F + F$, so the right side is $(F + F) \times \partial F \cong 2\,(F \times \partial F)$. The left side is, by Leibniz, $\partial F \times F + F \times \partial F \cong 2\,(F \times \partial F)$. They agree.*

Remark 2 (Faà di Bruno)**.** Iterating Theorem 1 together with Propositions 2 and 3 reproduces the Faà di Bruno formula for higher derivatives of a composite at container level; the partitions of Faà di Bruno are the ways of distributing holes among the inner blocks. We do not pursue this here.

Remark 3 (Computational verification)**.** The explicit pair (φ, ψ) was checked by brute force on 23 finite containers (one handcrafted family plus 20 randomly generated G, F with shape counts ≤ 3 and position counts ≤ 3, including empty position-sets): φ is a bijection of the (finitely many) shapes, and for every left shape ψ is a bijection onto the positions of φ of that shape. The polynomial invariant $\#[\![-]\!](n)$ agreed for $n \leq 5$.

5 Status and scope

- **Theorem 1 [Original].** The chain rule for the derivative under *composition/substitution* $\triangleleft$ is not in Abbott–Altenkirch–Ghani–McBride 2003 (which give the sum and product rules and a derivative *definition*, but not the $\triangleleft$ chain rule). Closest prior art: Joram–Veltri, arXiv:2512.17484 (Dec. 2024, Cubical Agda), who prove the chain rule in univalent foundations and show the classical (set-truncated) case holds — consistent with our Lemma 1. Cite them for the univalent case and for ∂'s definition alongside AAGM.

- **Derivation [MacBeth], container language.** Self-contained over **Set**; the only nonconstructive step is the splitting Lemma 1, flagged.

- **Lean [pending].** The product, coproduct and category structure on **Cont** are formalised (`Cont.lean`, PR #12); a Lean proof of Theorem 1 is the natural follow-on. The proof is finite and bijection-based, so it should transcribe directly; the dependent transports around $P(f\,q_0) = P\,s_*$ will need the `Ext.ext_eq` discipline noted for the D2/D5 laws.

- **Forward pointer (not chased here).** ∂ relates to zippers and to the cofree comonad; the one-hole context ∂F carries a comonad structure whose interaction with $\triangleleft$ deserves its own note.